

Root relationships- cubic, quartic

Basic skills

Q1. $z^2 + 10z - 4 = 0$

a) $\alpha + \beta = -\frac{b}{a} = -\frac{10}{1}$ $\alpha\beta = \frac{c}{a} = -\frac{4}{1}$

b) using

Sum: $1 - \alpha + 1 - \beta = 1 - (\alpha + \beta)$
 $= 1 - -10 = 11$

Product: $(1 - \alpha)(1 - \beta) = 1 - (\alpha + \beta) - \alpha\beta$
 $= 1 - -10 - -4$
 $= 15$

so $-\frac{b}{a} = 11$, $\frac{c}{a} = 15$

let $a = 1$ then $b = -11$, $c = 15$

so $z^2 - 11z + 15 = 0$

Could sub $z - (1 - \alpha) = 0$ $\alpha = 1 - z$

so $z \rightarrow 1 - z$

c) sum: $\frac{\alpha}{\beta} + \frac{\beta}{\alpha} = \frac{\alpha^2 + \beta^2}{\alpha\beta} = \frac{(\alpha + \beta)^2 - 2\alpha\beta}{\alpha\beta}$

$$= \frac{(-10)^2 - 2 \times -4}{-4} = \frac{108}{-4} = -27$$

Product: $\frac{\alpha}{\beta} \times \frac{\beta}{\alpha} = 1$

So $-\frac{b}{a} = -27, \quad \frac{c}{a} = 1$

let $a = 1$, then $b = 27, c = 1$

So $z^2 + 27z + 1 = 0$

Q2. $z^3 - z^2 + 10z - 4 = 0$

a) $\sum \alpha : \alpha + \beta + \gamma = -\frac{b}{a} = -\frac{-1}{1} = 1$

$\sum \alpha\beta : \alpha\beta + \alpha\gamma + \beta\gamma = \frac{c}{a} = \frac{10}{1} = 10$

$\sum \alpha\beta\gamma : \alpha\beta\gamma = -\frac{d}{a} = -\frac{-4}{1} = 4$

b) $\sum_{\text{new}} \alpha : (2\alpha + 1) + (2\beta + 1) + (2\gamma + 1) = 2(\alpha + \beta + \gamma) + 3$

$$= 2 \times 1 + 3 = 5$$

$$c) \sum_{\text{new}} \alpha\beta: (2\alpha+1)(2\beta+1) + (2\alpha+1)(28+1) \\ + (2\beta+1)(28+1)$$

$$= 4\alpha\beta + 2(\alpha+\beta)+1$$

$$+ 4\alpha 8 + 2(\alpha+8)+1$$

$$+ 4\beta 8 + 2(\beta+8)+1$$

$$= 4(\alpha\beta + \alpha 8 + \beta 8) + 4(\alpha + \beta + 8) + 3$$

$$= 4 \times 10 + 4 \times 1 + 3 = 47$$

$$d) \sum_{\text{new}} \alpha\beta 8: (2\alpha+1)(2\beta+1)(28+1) \\ = (4\alpha\beta + 2(\alpha+\beta)+1)(28+1) \\ = 8\alpha\beta 8 + 4\alpha\beta + 48(\alpha+\beta) + \\ 2(\alpha+\beta) + 28+1$$

$$= 8\alpha\beta 8 + 4(\alpha\beta + \alpha 8 + \beta 8) \\ + 2(\alpha + \beta + 8) + 1$$

$$= 8 \times 4 + 4 \times 10 + 2 \times 1 + 1$$

$$= 75$$

Competency skills

Q3. using Q2. $-\frac{b}{a} = 5$, $\frac{c}{a} = 47$, $-\frac{d}{a} = 75$

let $a = 1$, $b = -5$, $c = 47$, $d = -75$

$$\text{so } z^3 - 5z^2 + 47z - 75 = 0$$

Alternatively: by substitution

$$z - (2\alpha + 1) = 0 \quad \alpha = \frac{1}{2}(z-1)$$

$z \rightarrow \frac{1}{2}(z-1)$ in eq.

$$\left(\frac{1}{2}(z-1)\right)^3 - \left(\frac{1}{2}(z-1)\right)^2 + \frac{10}{2}(z-1) - 4 = 0$$

↓ binomial exp

$$\frac{1}{8}(z^3 - 3z^2 + 3z - 1) - \frac{1}{4}(z^2 - 2z + 1)$$

$$+ 5z - 5 - 4 = 0$$

$$z^3 - 3z^2 + 3z - 1 - 2z^2 + 4z - 2$$

$$+ 40z - 72 =$$

$$z^3 - 5z^2 + 47z - 75 = 0$$

Q4.

$$z^3 - 5z + 4 = 0$$

EXAM PAPERS PRACTICE

noticing $0z^2$

a) $\sum \alpha: \alpha + \beta + \gamma = -\frac{b}{a} = 0$

$$\sum \alpha\beta: \alpha\beta + \alpha\gamma + \beta\gamma = \frac{c}{a} = -\frac{5}{1} = -5$$

$$\sum \alpha\beta\gamma: \alpha\beta\gamma = -\frac{d}{a} = -\frac{4}{1} = -4$$

b) new

$$\begin{aligned} \sum \alpha: (3\alpha - 1) + (3\beta - 1) + (3\gamma - 1) \\ = 3(\alpha + \beta + \gamma) - 3 = 3 \times 0 - 3 = -3 \end{aligned}$$

new

$$\begin{aligned} \sum \alpha\beta: (3\alpha - 1)(3\beta - 1) + (3\alpha - 1)(3\gamma - 1) \\ + (3\beta - 1)(3\gamma - 1) \end{aligned}$$

$$= 9\alpha\beta - 3(\alpha + \beta) + 1$$

$$+ 9\alpha\gamma - 3(\alpha + \gamma) + 1$$

$$+ 9\beta\gamma - 3(\beta + \gamma) + 1$$

$$= 9 \sum \alpha\beta - 6 \sum \alpha + 3$$

$$= 9 \times -5 - 6 \times 0 + 3 = -45 + 3 = -42$$

new

$$\begin{aligned}\sum \alpha\beta\gamma &: (3\alpha-1)(3\beta-1)(3\gamma-1) \\ &= (9\alpha\beta - 3(\alpha+\beta) + 1)(3\gamma-1) \\ &= 27\alpha\beta\gamma - 9\alpha\beta - 9(\alpha\gamma+\beta\gamma) \\ &\quad + 3(\alpha+\beta) + 3\gamma - 1 \\ &= 27\sum \alpha\beta\gamma - 9\sum \alpha\beta + 3\sum \alpha - 1 \\ &= 27 \times -4 - 9 \times -5 + 3 \times 0 - 1 \\ &= -64\end{aligned}$$

$$\text{so } -\frac{b}{a} = -3, \quad \frac{c}{a} = -42, \quad -\frac{d}{a} = -64$$

$$\text{let } a=1, b=3, c=-42, d=64$$

$$\text{so } z^3 + 3z^2 - 42z + 64 = 0$$

Q5.

$$\begin{aligned}(\sum \alpha)^2 &= (\alpha+\beta+\gamma)(\alpha+\beta+\gamma) \\ &= \alpha^2 + \alpha\beta + \alpha\gamma + \alpha\beta + \beta^2 + \beta\gamma + \alpha\gamma \\ &\quad + \beta\gamma + \gamma^2 = \sum \alpha^2 + 2\sum \alpha\beta\end{aligned}$$

works similarly for quartic

Q6. $z^3 - 5z^2 + 6z + 1 = 0$

$$\sum \alpha = -\frac{b}{a} = -\frac{-5}{1} = 5, \quad \sum \alpha\beta = \frac{c}{a} = \frac{6}{1} = 6$$

$$\sum \alpha\beta\gamma = -\frac{d}{a} = -\frac{1}{1} = -1$$

a) new roots $\alpha\beta, \alpha\gamma, \beta\gamma$

$$\text{new } \sum \alpha: \alpha\beta + \alpha\gamma + \beta\gamma = \sum \alpha\beta = 6$$

$$\begin{aligned} \text{new } \sum \alpha\beta: & \alpha\beta \times \alpha\gamma + \alpha\beta \times \beta\gamma + \alpha\gamma \times \beta\gamma \\ &= \alpha^2\beta\gamma + \alpha\beta^2\gamma + \alpha\beta\gamma^2 \\ &= \alpha\beta\gamma(\alpha + \beta + \gamma) \\ &= \sum \alpha\beta\gamma \times \sum \alpha = -1 \times 5 = -5 \end{aligned}$$

$$\begin{aligned} \text{new } \sum \alpha\beta\gamma: & \alpha\beta \times \alpha\gamma \times \beta\gamma = \alpha^2\beta^2\gamma^2 \\ &= (\alpha\beta\gamma)^2 = (\sum \alpha\beta\gamma)^2 = (-1)^2 = 1 \end{aligned}$$

$$\text{so } -\frac{b}{a} = 6, \quad \frac{c}{a} = -5, \quad -\frac{d}{a} = 1$$

$$\text{let } a=1, b=-6, c=-5, d=-1$$

So $z^3 - 6z^2 - 9z - 1 = 0$

b) new roots $\alpha^2, \beta^2, \gamma^2$

new $\sum \alpha: \alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma)$ see Q5.

$$= 5^2 - 2 \times 6 = 13$$

new $\sum \alpha\beta = \alpha^2\beta^2 + \alpha^2\gamma^2 + \beta^2\gamma^2$

$$= (\alpha\beta)^2 + (\alpha\gamma)^2 + (\beta\gamma)^2$$

$$= (\alpha\beta + \alpha\gamma + \beta\gamma)^2 - 2(\alpha^2\beta\gamma + \alpha\beta^2\gamma + \alpha\beta\gamma^2)$$

$$= (\sum \alpha\beta)^2 - 2\alpha\beta\gamma(\alpha + \beta + \gamma)$$

$$= 6^2 - 2 \times -1 \times 5 = 46$$

new $\sum \alpha\beta\gamma: \alpha^2\beta^2\gamma^2 = (\alpha\beta\gamma)^2 = (-1)^2 = 1$

$$\text{so } -\frac{b}{a} = 13, \frac{c}{a} = 46, -\frac{d}{a} = 1$$

$$\text{let } a = 1, b = -13, c = 46, d = -1$$

$$\text{so } z^3 - 13z^2 + 46z - 1 = 0$$

Advanced skills

Q6.

old	Factors	new
$z - \alpha = 0$		$z - \alpha^2 = 0$
		$\sqrt{z} = \alpha$

so sub $z \rightarrow \sqrt{z}$ so NO!

However even following this fails
(non-linear substitutions are risky)

Q7. $z^4 - 3z^3 + 2z^2 - z + 2 = 0$

new root $2 - \alpha$

$$\text{so } z - (2 - \alpha) = 0 \quad \alpha = 2 - z$$

$z \rightarrow 2 - z$ so

$$(2 - z)^4 - 3(2 - z)^3 + 2(2 - z)^2 - (2 - z) + 2 = 0$$

$$16 - 4 \times 8z + 6 \times 4z^2 - 4 \times 2z^3 + z^4 \\ - 3(8 - 3 \times 4z + 3 \times 2z^2 - z^3) \\ + 2(4 - 4z + z^2) + z - 2 + 2 = 0$$

$$z^4 - 5z^3 + 8z^2 - 3z + 0 = 0$$

Q8. $2z^3 - 12z^2 + cz - 15 = 0$
 roots $\alpha = k, \beta = k\gamma, \gamma = k\gamma^2$

$$\sum \alpha \quad k + k\gamma + k\gamma^2 = -\frac{-12}{2} = 6$$

$$\sum \alpha\beta\gamma \quad k k\gamma k\gamma^2 = -\frac{-15}{2} = 7.5$$

$$k^3 \gamma^3 = 7.5 \quad k = \frac{\sqrt[3]{7.5}}{\gamma}$$

$$k + k\gamma + k\gamma^2 = 6$$

$$k(1 + \gamma + \gamma^2) = 6$$

$$\frac{\sqrt[3]{7.5}}{\gamma} (1 + \gamma + \gamma^2) = 6$$

$$\frac{1}{\gamma} + 1 + \gamma = \frac{6}{\sqrt[3]{7.5}}$$

$$1 + r + r^2 = \frac{6}{\sqrt[3]{2.5}} r$$

$$r^2 + r \left(1 - \frac{6}{\sqrt[3]{2.5}}\right) + 1 = 0$$

$$r = 1.29 \text{ or } 0.775$$

$$\text{So } k = 1.52 \text{ or } 2.53$$

So two options

$$\alpha = 1.29, \beta = 1.96, \gamma = 2.98$$

$$\text{or } \alpha = 0.775, \beta = 1.96, \gamma = 4.96$$

To find c

$$\sum \alpha\beta : \alpha\beta + \alpha\gamma + \beta\gamma$$

$$= 1.29 \times 1.96 + 1.29 \times 2.98 + 1.96 \times 2.98$$

$$= 12.21$$

or

$$15.08$$

TWO cubic have b, d
and geometric Progression

Q9.

$$z^4 + bz^3 + 30z + e = 0$$

$$\sum \alpha\beta : \alpha \times 5\alpha + \alpha \times \beta + \alpha \times -\beta \\ + 5\alpha \times \beta + 5\alpha \times -\beta + \beta \times -\beta$$

$$= 5\alpha^2 - \beta^2 \quad \text{so } 5\alpha^2 - \beta^2 = 0$$

$$\sum \alpha\beta\gamma : \alpha \times 5\alpha \times -\beta + \alpha \times 5\alpha \times \beta \\ \alpha \times \beta \times -\beta + 5\alpha \times \beta \times -\beta$$

$$= -6\alpha\beta^2 \quad \text{so } -6\alpha\beta^2 = -30$$

$$\text{so } \alpha\beta^2 = 5 \quad \text{and } \beta^2 = 5\alpha^2$$

$$5\alpha^3 = 5 \quad \alpha = 1 \quad \text{so } \beta = \sqrt{5}$$

$$\text{Then } \frac{-b}{a} = \alpha + 5\alpha + \beta - \beta$$

$$\text{so } b = -6\alpha = -6$$

$$\frac{e}{a} = \alpha \times 5\alpha \times \beta \times -\beta = -5\alpha^2\beta^2$$

$$e = -5 \times 1^2 \times (\sqrt{5})^2 = -25$$

Q10. $az^3 + bz^2 + cz + d = 0$

Let roots be $\alpha, \frac{1}{\alpha}, \beta$

Then

$$\sum \alpha: \alpha + \frac{1}{\alpha} + \beta = -\frac{b}{a}$$

$$\sum \alpha\beta: \alpha \times \frac{1}{\alpha} + \alpha \times \beta + \frac{1}{\alpha} \times \beta = \frac{c}{a}$$

$$\sum \alpha\beta\gamma: \alpha \times \frac{1}{\alpha} \times \beta = -\frac{d}{a}$$

$$\text{So } \beta = -\frac{d}{a}, \quad \alpha + \frac{1}{\alpha} + \beta = -\frac{b}{a}$$

$$1 + \alpha\beta + \frac{\beta}{\alpha} = \frac{c}{a}$$

$$\alpha + \frac{1}{\alpha} = -\frac{b}{a} + \frac{d}{a}, \quad 1 + -\frac{d\alpha}{a} - \frac{d}{a\alpha} = \frac{c}{a}$$

$$a - d\alpha - \frac{d}{\alpha} = c$$

$$-d\left(x + \frac{1}{x}\right) = c - a$$

$$x + \frac{1}{x} = \frac{a - c}{d}$$

$$\text{So } -\frac{b}{a} + \frac{d}{a} = \frac{a - c}{d}$$

$$-bd + d^2 = a^2 - ac$$

$$d^2 - a^2 = bd - ac$$